

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

*Artificial Intelligence
and Society*

Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 4: LLMs and Recommender Systems

Jana Lasser

Research group

Complex Social & Computational Systems (CS)²

IDEa_Lab

Das Interdisziplinäre Digitale
Labor der Universität Graz



Review Lecture 3

- **Question:** *How does unsupervised learning work and what distinguishes it from supervised learning?*

Answer: Unsupervised learning, also known as cluster analysis, tries to find unknown patterns in data. Unlike supervised learning, there are no predefined classes or target values for observations.

- **Question:** *Why are distance or similarity measures important for unsupervised learning?*

Answer: We need distance or similarity measures to determine how "close" or "far apart" two data points are from each other. Based on this, we can find clusters in the data.

- **Question:** *How does the k -Means algorithm work?*

Answer: Starting from a given clustering, data points are moved to other clusters until the clustering is optimal (the sum of squared errors is minimal).

Overview of Today's Lecture

- 4.1 Text as Data
- 4.2 Sequence Prediction
- 4.3 Generative Language Models
- 4.4 Intro to Recommender Systems
- 4.5 Designing Recommender Systems
- 4.6 Recommender System of X

Paper [word2vec Parameter Learning Explained](#)

Video [CS 198-126: Lecture 14 - Transformers and Attention](#)

Paper [Language Models are Few-Shot Learners](#)

Lecture Slides [Recommender Systems: Value, Methods, Measurements](#)

Blog Post [Twitter's Recommendation Algorithm](#)

[content for further reading]

Learning Objectives

- Understand how large language models are structured and why they hallucinate and make mistakes.
- Understand how recommender systems are structured and how they relate to societal problems.